

Risk Assessment Record		Project: Soapamorph				<i>To be assessed yearly</i>	
Risk Assessed By:	Rachel Kirkham						
Name:	Group A	Position:	Student	Location:	Mellor Building.		
Date:	22/02/2026	Level:	6	Department:	Department of Computing		
Risk Matrix – Use this to determine level of risk for each hazard i.e. ‘How bad & How likely’							
Severity of Harm	(1) Rare (e.g. once every few years)	(2) Unlikely (e.g. Once per year)	(3) Possible (e.g. Monthly)	(4) Likely (e.g. Weekly)	(5) Very Likely (e.g. daily or frequent)		
(1) Negligible (e.g. Slight Bruising)	Low (1)	Low (2)	Low (3)	Medium (4)	Medium (5)		
(2) Minor (e.g. Small cuts, deep bruising)	Low (2)	Low (4)	Medium (6)	Medium (8)	Medium (10)		
(3) Moderate (e.g. Deep cuts, Torn muscles)	Low (3)	Medium (6)	Medium (9)	High (12)	High (15)		
(4) Severe (e.g. Fracture, Loss of consciousness)	Medium (4)	Medium (8)	High (12)	High (16)	Extremely High (20)		

(5) Very Severe (e.g. Death, Permanent disability/disease)	Medium (5)	High (10)	High (15)	Extremely High (20)	Extremely High (25)			
Legend:	Low (1-4)		Medium (4-10)	High (11-15)	Extreme (16-25)			
Hazard List – List below.								
Hazards identified	Risk	Potential injuries/ damage	Likelihood (1-5)	Severity (1-5)	Risk Rating (Likelihood x Severity)	Who is at risk	Additional measures/ Precautions made?	Who is responsible for action?
Electrical or battery fault	- Chemical Leak/ burns - Fire/ overheating - Electric shock - Fire	- Chemical burns - Laboratory damage - Electric shock - Shock - Burns	2	4	8 (Medium)	Team members	- All electrical equipment will have been PAT tested. - Pre-use inspection. - Switch on main power.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Animatronic Malfunction	- Unexpected or uncontrolled movement - Minor injury	Bruising	2	1	2 (Low)	Team members	- Switch on main power. - Plenty of space to stand away from project.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Obstacles / Cluttered Area	- Trip/slip hazard - Collison - Blockage of exits	- Slip/ Trip - Injury to back - Sprain/ break - Head injury - Smoke inhalation - Blocked in fire	2	4	8 (Medium)	Team members	- There will be plenty of manoeuvrable space. - Exits and walkways kept clear.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller

		Minor injuries from cluttered workspace (such as dropping onto feet)					- Storage locations for belongings. - Workplace kept tidy.	
Objects falling	- Heavy items falling - Tools falling.	- Breakages - Bruising - Fracture	1	4	4 (Low)	Team members	- Tidy workspace. No cluttered surfaces. - Storage area for belongings. - Appropriate use of PPE - steel toe caps.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Cutting, welding metal.	- Burns. - Sparks. - Foreign objects. - Contact with equipment. - Bright/piercing lights. - Snagged clothing. - Untidy workplace. - Falling objects. - Damaged equipment.	- Blindness. - Eye injuries. - Severe burns. - Small cuts. - Deep cuts. - Injured limbs. - Damage to feet. - Long term damage.	2	5	10 (Medium)	Team members	- Wear appropriate PPE (Gloves, Welding mask, Goggles) - Wear appropriate tight clothing - tight fitted. - Wear steel toe cap safety boots. - Keep workstation tidy and all exits/ walkways tidy. - All equipment checked before and after use for wear and tear. - All equipment checked for PAT testing.	Rachel Kirkham.

Soldering.	• Contact with equipment. • Fire. • Untidy workplace. • Falling objects. • Fumes. • Foreign objects/ Objects in eye. • Damaged equipment.	• Eye injuries. • Severe burns. • Lung disease/ injury. • Electrocution. • Shock.	2	5	10 (Medium)	Team members	• Wearing of appropriate PPE (goggles and steel toe caps). • Wearing appropriate tight fitted clothing. • Stand up during soldering process to limit risk of solder dripping onto legs. • All equipment checked for wear and tear before and after use. • All equipment checked for PAT testing.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Silicone moulding	• Toxic fumes. • Chemical exposure. • Poor ventilation. • Heat resistance failures. • Incomplete curing. • Incorrect manual handling. • Sharp tools and equipment. • Spills.	• Skin irritation. • Chemical burns. • Allergic reactions. • Respiratory irritation. • Headaches. • Dizziness. • Nausea. • Skin blistering. • Minor to Severe burns. • Muscle or back strain.	2	4	8 (Medium)	Team members	• Wear appropriate PPE, gloves, goggles, lab coat, steel toe cap boots etc.... • Keep a well-ventilated area. • Heat sources supervised. • Keep workspace tidy and clean any spillages.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller

	Cluttered work area.	- Cuts/punctures. - Slips and falls. - Falling objects.					- Wear gloves for sharp tool exposure. - Follow COSHH manufacturer data sheet. - Wash hands before and after (don't touch face during process).	
Laser cutting	- Fire. - Poor ventilation. - Exposure to laser lighting. - Moving parts/mechanical injury/malfunctions. - Damages to equipment. - Damages to wiring or power source. - Sharp edges/broken material handling.	- Smoke/ toxic fume inhalation. - Eye strain to permanent damage/blindness. - Cuts/punctures. - Electrical shock. - Bruising. - Minor to major muscle injury. - Respiratory irritation/long term damage. - Headaches. - Dizziness. - Nausea.	1	4	4 (Low)	Team members	- Wear appropriate PPE, glasses if needed. - Make sure ventilation on machine works. - Check machine for PAT testing. - Check machine for external and internal damages. - Check wiring to power socket. - Make sure emergency button is easily nearby. - Do not get material out of machine for 20 seconds to 1 minute to allow	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller

							cooling and ventilation.	
3D printing	- Plastic fumes. - Fire. - Poor ventilation. - Heat buildup. - Malfunctions. - Exposed/damaged wiring. - Damages to machine present.	- Respiratory irritation/long term damage. - Headaches. - Dizziness. - Nausea. - Electrical shock. - Smoke/ toxic fume inhalation.	1	4	4 (Low)	Team members	- Check machine ventilation. - Check machine has been PAT tested. - Check internal and external parts of machine for signs of damage before and after use. - Check wiring and main power connection for signs of damage.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Fusion/blender	- Screen strain. - Incorrect posture long term.	- Eye strain. - Back / muscle strain.	2	2	4 (Low)	Team members	Take consistent breaks throughout.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Assessor Signature:	<i>R. Kirkham</i>							
Date:	22/02/2026							